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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Music Store Data Analysis

A SQL-Driven Analysis of Sales, Customers & Music Trends

Presented By:

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ABOUT ME

NAME :Dornala Vishnu vardhan reddy

Background: B.Tech – IT(INFORMATION TECHNOLOGY)

Why Data Science? Passionate about solving real-world problems using data and building analytical solutions.

Work Experience: Fresher

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Objective :

- Design and implement a Music Store relational database using MySQL.
- Analyze customer, sales, track, artist, album, and genre data to identify meaningful business insights.
- Use advanced SQL techniques such as JOINs, aggregations, subqueries, CTEs, and window functions to answer real-world business questions.
- Identify insights related to customer spending, sales performance, popular genres, artists, and purchasing patterns to support data-driven decision-making.

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Problem Statement :

- Managing music store data across multiple related tables can make it difficult to obtain meaningful business insights.

- Customer purchases, invoices, tracks, artists, albums, and genres need to be analyzed together to understand purchasing behavior.

- It is challenging to identify top customers, high-revenue cities, popular genres, leading artists, and country-wise purchasing patterns from raw data.

- A centralized relational database and SQL-based analysis are required to efficiently organize the data and generate actionable insights for better business decision-making.

Overview :

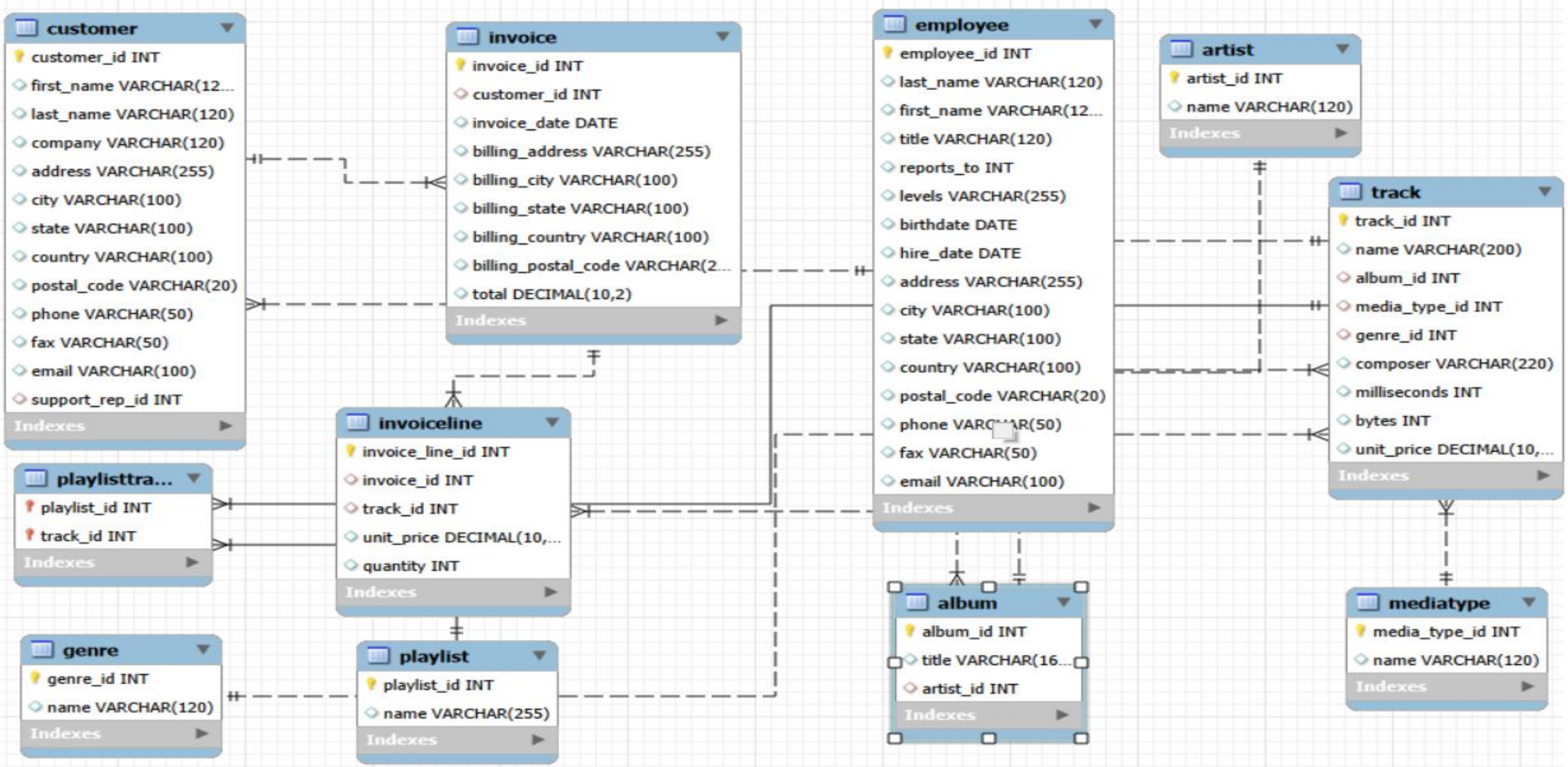
Technologies Used :

- MySQL
- SQL
- MySQL Workbench

Key Features :

- Relational Database Design
- CSV Data Import & Management
- Customer & Sales Analysis
- Artist & Genre Analysis
- Country-wise Purchase Analysis
- Advanced SQL Queries
- Business Insight Generation

ER Diagram :



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1. Employee & Business Analysis:

Q. Who is the senior most employee based on job title?

Query :

```
SELECT * FROM employee;
SELECT first_name,
last_name,
title,
hire_date
FROM employee
ORDER BY hire_date ASC
LIMIT 1;
```

Output :

Result Grid | Filter Rows: | Export: | W

first_name	last_name	title	hire_date
Mohan	Madan	Senior General Manager	2016-01-14

Key Insights :

- **Mohan Madan**, Senior General Manager, is the **senior-most employee** based on the earliest hire date.
- He joined the organization on **14 January 2016**, making him the longest-serving employee in the dataset.
- The result demonstrates how **sorting by hire date** can be used to identify employee seniority.

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2. Sales & Revenue Analysis:

Q. Which countries have the most Invoices?

Query :

```
SELECT billing_country,
COUNT(invoice_id) AS TOTAL_INVOICES
FROM Invoice
GROUP BY billing_country
ORDER BY TOTAL_INVOICES DESC;
```

Output :

billing_country	TOTAL_INVOICES
USA	131
Canada	76
Brazil	61
France	50
Germany	41
Czech Republic	30
Portugal	29
United Kingdom	28
India	21
Ireland	13
Chile	13
Finland	11
Spain	11
Poland	10
Denmark	10

Key Insights :

- The analysis identifies the **countries with the highest number of invoices**.
- Countries with a higher invoice count represent **greater customer purchase activity**.
- Invoice distribution across countries helps identify **key geographic markets** for the music store.
- These markets can be targeted for **regional promotions and customer engagement strategies**.

3. Sales & Revenue Analysis:

Q. What are the top 3 values of total invoice?

Query :

```
SELECT invoice_id,
    total
FROM invoice
ORDER BY total DESC
LIMIT 3;
```

Output :

invoice_id	total
183	23.76
92	19.80
31	19.80
NULL	NULL

Key Insights :

- The highest invoice value is **$23.76**.
- The second and third highest invoice values are both **$19.80**, indicating a tie.
- The top invoices highlight the **highest-value individual transactions** in the store.

4.Sales & Revenue Analysis:

Q. Which city has the best customers? - We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals

Query :

```
SELECT billing_city,
    SUM(total) AS total_sales
FROM Invoice
GROUP BY billing_city
ORDER BY total_sales DESC
LIMIT 1;
```

Output :

billing_city	total_sales
Prague	273.24

Key Insights :

- **Prague** generated the highest total invoice revenue of **$273.24**
- It is the strongest-performing city based on total customer purchases.
- Prague would be an ideal location for the proposed **promotional Music Festival**..

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5. Sales & Revenue Analysis

Q. Who is the best customer? - The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money

Query :

```
SELECT c.customer_id,
    c.first_name,
    c.last_name,
SUM(i.total) AS total_spent
FROM customer c
JOIN invoice i
    ON c.customer_id = i.customer_id
GROUP BY c.customer_id,
    c.first_name,
    c.last_name
ORDER BY total_spent DESC
LIMIT 1;
```

Output :

customer_id	first_name	last_name	total_spent
5	FrantiÅ¡ek	WichterlovÃ¡	144.54

Key Insights :

- **František Wichterlová** is the highest-spending customer, with total purchases of **$144.54**.
- The result identifies the customer's **highest-value contributor to store revenue**.
- High-spending customers can be targeted with **loyalty rewards and personalized promotions**.

6. Customer & Genre Analysis

Q. Write a query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A

Query :

```sql
SELECT DISTINCT
    c.email,
    c.first_name,
    c.last_name,
    g.name AS genre
FROM Customer c
JOIN Invoice i
    ON c.customer_id = i.customer_id
JOIN InvoiceLine il
    ON i.invoice_id = il.invoice_id
JOIN Track t
    ON il.track_id = t.track_id
JOIN Genre g
    ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
ORDER BY c.email ASC;
```

Output :

email	first_name	last_name	genre
aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock
alero@uol.com.br	Alexandre	Rocha	Rock
astrid.gruber@apple.at	Astrid	Gruber	Rock
bjorn.hansen@yahoo.no	BjÃ¸rn	Hansen	Rock
camille.bernard@yahoo.fr	Camille	Bernard	Rock
daan_peeters@apple.be	Daan	Peeters	Rock
diego.gutierrez@yahoo.ar	Diego	GutiÃ©rrez	Rock
dmiller@comcast.com	Dan	Miller	Rock
dominiquelefebvre@gmail.com	Dominique	Lefebvre	Rock
edfrancis@yachoo.ca	Edward	Francis	Rock

Key Insights :

- The analysis identifies customers who have purchased **Rock music**.
- The customer list contains **59 unique Rock listeners** in the dataset.
- Ordering by email alphabetically makes the customer list **easy to search and reference**.
- Rock demonstrates a strong presence among the store's purchased music.

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7. Artist & Track Analysis

Q. Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands

Query :

```
SELECT
    ar.name AS artist_name,
    COUNT(t.track_id) AS total_tracks
FROM Artist ar
JOIN Album al
    ON ar.artist_id = al.artist_id
JOIN Track t
    ON al.album_id = t.album_id
JOIN Genre g
    ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY ar.artist_id, ar.name
ORDER BY total_tracks DESC
LIMIT 10;
```

Output :

artist_name	total_tracks
AC/DC	18
Aerosmith	15
Audioslave	14
Led Zeppelin	14
Alanis Morissette	13
Alice In Chains	12
Frank Zappa & Captain Beefheart	9
Accept	4

Key Insights :

- **AC/DC** has the highest number of Rock tracks with **18 tracks**.
- **Aerosmith** and **Audioslave** follow with **15 and 14 tracks**, respectively.
- The analysis identified **8 Rock artists** in the dataset, despite requesting the top 10.

8. Artist & Track Analysis

Q. Return all the track names that have a song length longer than the average song length.- Return the Name and Milliseconds for each track. Order by the song length, with the longest songs listed first

Query :

```
SELECT name,
    milliseconds
FROM track
where milliseconds >(
    SELECT AVG(milliseconds)
    FROM track)
ORDER BY milliseconds DESC;
```

Output :

name	milliseconds
How Many More Times	711836
Advance Romance	677694
Sleeping Village	644571
You Shook Me(2)	619467
Talkin' 'Bout Women Obviously	589531
Stratus	582086
No More Tears	555075
The Alchemist	509413
Wheels Of Confusion / The Straightener	494524
Book Of Thel	494393
You Oughta Know (Alternate)	491885

Key Insights :

- **"How Many More Times"** is the longest track returned by the analysis, with **711,836 milliseconds (~11.86 minutes)**.
- **"Advance Romance"** and **"Sleeping Village"** follow with **677,694 ms** and **644,571 ms**, respectively.
- The analysis identifies the **longest tracks that exceed the average song length**, helping highlight extended recordings in the music catalog.

9. Customer–Artist Spending Analysis

Q. Find how much amount is spent by each customer on artists? Write a query to return customer name, artist name and total spent

Query :

```
SELECT
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    ar.name AS artist_name,
    SUM(il.unit_price * il.quantity) AS total_spent
FROM Customer c
JOIN Invoice i
    ON c.customer_id = i.customer_id
JOIN InvoiceLine il
    ON i.invoice_id = il.invoice_id
JOIN Track t
    ON il.track_id = t.track_id
JOIN Album al
    ON t.album_id = al.album_id
JOIN Artist ar
    ON al.artist_id = ar.artist_id
GROUP BY
    c.customer_id,
    c.first_name,
    c.last_name,
    ar.artist_id,
```

Output :

customer_name	artist_name	total_spent
Steve Murray	AC/DC	17.82
Jennifer Peterson	Aerosmith	14.85
Mark Taylor	Aerosmith	14.85
Leonie KÃ¶hler	Audioslave	13.86
Fernanda Ramos	AntÃ´nio Carlos Jobim	13.86
Edward Francis	Alanis Morissette	12.87
Emma Jones	Alanis Morissette	12.87
JoÃ£o Fernandes	Alanis Morissette	12.87
Victor Stevens	Alice In Chains	11.88
Phil Hughes	AC/DC	10.89

Key Insights :

- Steve Murray spent the highest amount on a single artist, AC/DC, with $17.82.
- **Jennifer Peterson and Mark Taylor** both spent **$14.85 on Aerosmith**, showing a tie in customer–artist spending.
- The analysis highlights **customer spending preferences across different artists**.

10. Country-wise Music Analysis

Q. We want to find out the most popular music Genre for each country.We determine the most popular genre as the genre with the highest amount of purchases.Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared, return all Genres**Query :**

```sql
WITH GenrePurchases AS (
    SELECT
        i.billing_country AS country,
        g.name AS genre,
        COUNT(il.invoice_line_id) AS total_purchases
    FROM Customer c
    JOIN Invoice i
        ON c.customer_id = i.customer_id
    JOIN InvoiceLine il
        ON i.invoice_id = il.invoice_id
    JOIN Track t
        ON il.track_id = t.track_id
    JOIN Genre g
        ON t.genre_id = g.genre_id
    GROUP BY
        i.billing_country,
        g.name
),
RankedGenres AS (
    SELECT
        country,
        genre,
        total_purchases,
        RANK() OVER (
            PARTITION BY country
            ORDER BY total_purchases DESC
        ) AS genre_rank
    FROM GenrePurchases
)
SELECT
    country,
    genre,
    total_purchases
FROM RankedGenres
WHERE genre_rank = 1
ORDER BY country;
```

Output :

country	genre	total_purchases
Argentina	Rock	1
Australia	Rock	18
Austria	Rock	6
Belgium	Rock	5
Brazil	Rock	26
Canada	Rock	57
Chile	Rock	7
Czech Republic	Rock	14
Denmark	Rock	6
Finland	Rock	6
France	Rock	26
Germany	Rock	28
Hungary	Rock	4
India	Rock	13

Key Insights :

- **Rock is the top genre across the countries.**
- **USA has the highest number of Rock purchases with 70**, followed by **Canada with 57**.
- **Germany has 28**, while **Brazil and France each have 26** Rock purchases.

11. Country-wise Music Analysis

Q. Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount

Query :

```sql
WITH CustomerSpending AS (
    SELECT
        i.billing_country AS country,
        c.customer_id,
        c.first_name,
        c.last_name,
        SUM(i.total) AS total_spent
    FROM Customer c
    JOIN Invoice i
        ON c.customer_id = i.customer_id
    GROUP BY
        i.billing_country,
        c.customer_id,
        c.first_name,
        c.last_name
),
RankedCustomers AS (
    SELECT
        country,
        customer_id,
        first_name,
        last_name,
        total_spent,
        RANK() OVER (
            PARTITION BY country
            ORDER BY total_spent DESC
        ) AS customer_rank
    FROM CustomerSpending
)
SELECT
    country,
    first_name,
    last_name,
    total_spent
FROM RankedCustomers
WHERE customer_rank = 1
ORDER BY country;
```

Output :

country	first_name	last_name	total_spent
Argentina	Diego	GutiÃ©rrez	39.60
Australia	Mark	Taylor	81.18
Austria	Astrid	Gruber	69.30
Belgium	Daan	Peeters	60.39
Brazil	LuÃ-s	GonÃ§alves	108.90
Canada	FranÃ§ois	Tremblay	99.99
Chile	Luis	Rojas	97.02
Czech Republic	FrantiÅiek	WichterlovÃi	144.54
Denmark	Kara	Nielsen	37.62
Finland	Terhi	HÃ¤mÃ¤lÃ¤inen	79.20
France	Wyatt	Girard	99.99
Germany	Fynn	Zimmermann	94.05
Hungary	Ladislav	KovÃics	78.21
India	Manoj	Pareek	111.87
Ireland	Hugh	O'Reilly	114.84

Key Insights :

- **František Wichterlová** is the highest-spending customer in the results, spending **$144.54** in the the **Czech Republic**.
- **Hugh O'Reilly ($114.84)** and **Manoj Pareek ($111.87)** are among the other highest-spending customers in their respective countries.

Challenges Faced :

- Understanding and mapping relationships across 10 interrelated tables, especially the multi-hop chain connecting **Customer → Invoice → InvoiceLine → Track → Genre/Artist**.
- Handling large data volumes while importing **track.csv, invoice_line.csv**, and **playlist_track.csv**, which took longer to load and required patience during the Table Data Import Wizard process.
- Writing multi-level JOIN queries to connect tables that had no direct foreign key relationship (e.g., linking Customers to Genres through three intermediate tables).
- Correctly identifying business logic behind each question — for example, distinguishing "senior most employee by title" (hierarchy-based, using reports_to) from "senior most by tenure" (date-based, using hire_date).
- Handling ties in ranking questions (e.g., top genre or top customer per country) using **RANK()/DENSE_RANK()** instead of a simple **LIMIT**, to ensure all tied results were returned as required.
- Ensuring foreign key constraints were respected during import — importing child tables (like Album) before their parent tables (like Artist) caused rows to silently fail.
- Formatting and validating date fields consistently to support accurate time-based and trend analysis.

Conclusion :

- Successfully designed and implemented a Music Store Data Analysis system using SQL, based on a 10-table relational schema (Chinook-style database).
- Built a fully normalized database with proper Primary Keys and Foreign Keys connecting Customers, Employees, Invoices, Tracks, Albums, Artists, Genres, and Playlists.
- Performed in-depth SQL analysis covering customer behavior, sales trends, top-performing artists and genres, and revenue insights by country and city.
- Applied advanced SQL techniques including multi-table JOINs, aggregations, subqueries, and window functions to handle ranking and tie-breaking scenarios.
- Delivered actionable business insights — such as identifying the best customer, the most profitable city, and the most popular genre per country — demonstrating how SQL can directly support real business decisions.
- Demonstrated the ability to translate a business question into an efficient, accurate SQL query — a core skill for a Data Analyst role.

THANK YOU

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